

FIVE GUYS × NEWSVENDOR

A sigma-reduction strategy for the European QSR outlier. Written companion to the 15-04-26 presentation deck.

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01 THE COMPANY

REAL · VISIBLE · CONSEQUENTIAL

Five Guys is the operational outlier in European QSR. 294 stores across the UK, France, Spain and Germany, a 25% ten-year sales CAGR, a 22% ten-year store CAGR, and zero above-the-line advertising. Full European potential is ~870-950 stores — roughly a 3x runway from here (*Deck, Slide 01*). The constraints that make the food great — no freezers, no timers, no shortcuts, fresh same-day prep with fifteen-plus free toppings — also make the operations hard. Every store runs, in effect, a daily newsvendor bet with no salvage value.

Disney Village Paris is the busiest Five Guys location in France and the sharpest instance of the problem: peak demand of ~€4,000/hour, a four-hour park visit window during which guests are not eating, and a forecast that has to absorb weather, ride closures, school holidays, and concert schedules at the adjacent venue. We chose the site because the operations problem is real and visible there in a way it is not on a high-street location. A full team site visit on 08-04-26 with District Manager Stephanie Barth (four stores, 280 staff) provided the field data underpinning this report.

02 DIAGNOSIS

ONE NUMBER RUNS THE STORE

Every morning each store is handed a single forecasted revenue number. At Disney Village on the day of our visit, that number was **€28,000**. Everything downstream — patties, potatoes, lettuce, labour — is a deterministic conversion from that number, mediated by an Excel "prep box" formula that takes the daily target plus current fridge inventory and outputs the day's prep quantities. Nobody on the floor could interrogate the €28,000. This is the structural problem the project addresses.

We did not have access to Five Guys' internal POS dataset, so the working below mixes *observed* values from the site visit with *plausible estimates* calibrated to what the District Manager shared. The epistemic status of each parameter is stated in the block.

NEWSVENDOR WORKING · DISNEY VILLAGE · 08-04-26

μ = **€28,000** · *observed* — the morning forecast handed to the store on the day of our visit

$C_u / (C_u + C_o)$ $\approx$ **0.99** · *inferred* — underage is catastrophic ("not allowed to be open if can't serve burgers or fries"); overage is absorbed within short shift windows (meat 3 shifts, veg 4). The ratio approaches 1.

σ $\approx$ **±€7,000** · *estimated* — calibrated to observed forecast-vs-actual dispersion (one extreme day: forecast €40k / actual €15k) and the chain waste line (€187k weekly turnover, 1.8% waste = €3,366, front-loaded into daily over-prep)

Q^* $\approx$ **€50,000** *prep* · *derived* — rational newsvendor response at the implied ~99% service level given μ and σ above

Observed: μ and the waste line. Inferred / estimated: $C_u / (C_u + C_o)$ and σ . Derived: Q^* . Sharper calibration would require POS data across stores

03 TECHNOLOGY INTERVENTION

YOU DON'T NEED A BETTER MODEL. YOU NEED A TOURNAMENT.

The asset that makes the intervention possible is not a model but a dataset. Ten years $\times$ 294 stores $\approx$ **one million labelled store-days** of forecast-vs-actual pairs, generated every day and — once consolidated across Europe — a uniquely rich training set for a generative AI forecasting system. The opportunity is not to build a better forecast. It is to run a permanent competition between many forecasts and let the data pick the winner, per store, per day.

The redesigned process is a three-stage tournament (*Deck, Slide 03*). **Stage 1:** hundreds of candidate models are trained on signals the current manual forecast cannot absorb — Disney park attendance, weather, school holidays, hotel occupancy, local events, lagged POS, football fixtures, public holidays, day-of-week seasonality, concert schedules, competitor proximity. **Stage 2:** every model makes a prediction; every prediction is scored against actual revenue at close of business. Every store, every day. **Stage 3:** winners are promoted, losers retired. Models are selected per store and per cluster — Disney Village learns to weight park attendance; London City learns to weight office-worker density. The system becomes a self-refreshing forecasting capability, not a one-off model build.

The POM mechanism is a demand-side real option analogous to the pre-sales information lever (*Session 07*). Same Cu, same Co, same critical ratio — tighter Q^* , lower expected cost on every day, not just on the tails.

ILLUSTRATIVE IMPACT · PER STORE · TOURNAMENT Σ TARGET VS CURRENT

	TODAY (EST.)	WITH TOURNAMENT
Forecast σ (daily)	$\sim \pm \text{€}7,000$	$\sim \pm \text{€}3,000$
Q^* at $\sim 99\%$ service	$\sim \text{€}50,400$	$\sim \text{€}37,600$
Prep overshoot vs mean	$\sim +80\%$	$\sim +34\%$
Per-store daily saving	—	$\sim \text{€}12,800$

Target $\sigma = \pm \text{€}3,000$ is indicative, benchmarked against typical gains reported in QSR demand-forecasting case studies. Actual uplift would be measured out-of-sample in pilot. The purpose of the table is to size the prize, not to promise a specific outcome.

$\text{€}12.8\text{K} / \text{DAY} \times 294 \text{ STORES} \times 50\% \text{ REALISM DISCOUNT} = \text{€}50\text{-}200\text{M}$
INDICATIVE OPERATIONAL LEVER · NO CAPEX

04 HONEST RISKS & WHAT MUST BE TRUE

THEORY VS REALITY

NEW RISKS THE INTERVENTION CREATES

Tail events are thinnest in history — the Frozen launch, a park closure, a strike. Model confidence is weakest precisely where the cost is highest. Mitigation: explicit human-in-the-loop override on days flagged as out-of-distribution, with each override logged as training signal.

False precision. A tighter σ that is systematically wrong is worse than an honest wide σ . Calibration must be monitored continuously, not annually.

Forecast $\neq$ full bottleneck. Stephanie's real-time judgement on staff allocation, station rebalancing and kiosk failures sits downstream of σ . A tournament improves the input she works from; it does not replace her decisions on the day.

Organisational trust. District Managers told us they do not auto-order today because "the difference between theory and reality is significant." The system must earn the right to reduce Q^* progressively, not impose it.

WHAT MUST BE TRUE · ORG · DATA · IMPLEMENTATION

POS feeds centralised across 294 stores. Currently store-level, minimally networked. Without this, no tournament.

Forecast transparency. Every output shows drivers and confidence, so a District Manager can interrogate the number in a way she today cannot.

Override as training signal. Manager corrections feed back into the model registry. Agent advises; manager decides; the system learns from the disagreement.

Pilot-then-scale. Start with the 38 French stores, Disney Village as flagship. Scale to 294 once in-store calibration and trust are demonstrated.

Realistic time horizon. 18-24 months to production; longer to organisational trust — consistent with the UPS ORION lesson that implementation is a decade-long journey.